

PITTSFIELD-PENSTONE, IL, USA

WMO: 744663

Lat: 39.639N

Lon: 90.778W

Elev: 216

StdP: 98.75

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-13.1	-21.3	0.6	-15.1	-18.8	0.7	-12.2	10.9	-2.1	9.8	-0.1	2.6	290	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.0	24.2	32.6	24.1	31.4	23.5	26.6	31.1	25.6	30.4	24.9	29.6	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.9	29.1	24.1	19.5	28.4	23.2	18.5	27.5	84.3	31.2	80.0	30.6	76.5	29.7	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.2	7.4	DB	-20.5	36.5	2.8	2.1	-22.5	38.0	-24.2	39.2	-25.8	40.4	-27.9	41.9	(4)
(5)			WB	-20.7	28.0	2.8	0.8	-22.7	28.6	-24.4	29.1	-26.0	29.5	-28.0	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	-1.9	-0.5	6.6	12.8	18.4	22.8	24.7	24.1	20.5	13.0	6.7	0.8	(6)	
	DBStd	10.74	6.70	6.52	6.32	5.36	4.80	3.33	2.94	3.04	4.15	5.37	5.46	5.59	(7)	
	HDD10.0	1294	372	300	144	31	2	0	0	0	29	127	289		(8)	
	HDD18.3	2867	628	529	366	179	62	5	1	24	180	349	544		(9)	
	CDD10.0	2169	2	5	38	114	263	384	456	438	314	124	28	3	(10)	
	CDD18.3	699	0	0	2	12	64	139	198	180	88	16	1	0	(11)	
	CDH23.3	6573	0	0	16	129	512	1236	2009	1687	835	143	6	0	(12)	
(13)	CDH26.7	2446	0	0	1	26	137	435	821	682	309	36	1	0	(13)	
(14)	Wind	WSAvg	3.1	3.7	3.7	3.9	3.9	3.2	2.6	2.0	1.8	2.2	2.9	3.5	3.5	(14)
(15) Precipitation	PrecAvg	935	41	38	78	90	103	96	103	82	91	75	69	59	(15)	
	PrecMax	1325	131	86	182	197	268	204	411	206	262	184	242	190	(16)	
	PrecMin	601	2	0	16	16	11	32	8	21	5	11	12	4	(17)	
	PrecStd	190	31	20	39	48	50	51	81	46	56	41	47	34	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	19.8	25.6	29.6	31.8	33.8	36.8	37.1	34.3	30.3	23.7	18.0	(19)	
		MCWB	13.3	13.3	16.8	17.7	22.3	23.6	24.0	24.0	22.4	21.5	15.7	15.0	(20)	
	2%	DB	13.1	16.0	22.3	27.0	29.9	32.4	33.7	34.0	32.3	27.2	20.8	14.5	(21)	
		MCWB	10.1	11.6	15.3	16.4	21.1	23.9	25.4	24.0	22.9	18.8	14.9	12.0	(22)	
	5%	DB	10.3	12.5	19.0	24.2	28.0	31.1	32.4	32.4	30.6	24.2	17.8	11.6	(23)	
		MCWB	7.1	8.9	13.9	15.6	20.0	23.6	24.9	23.9	22.3	17.3	13.3	8.9	(24)	
	10%	DB	7.3	9.0	16.4	22.2	26.4	29.2	31.1	30.8	27.9	22.1	15.3	8.8	(25)	
		MCWB	5.0	6.1	12.0	14.8	19.1	22.6	24.4	23.0	20.4	16.8	11.2	6.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.5	18.5	19.7	24.1	26.2	28.0	27.7	25.7	23.1	18.0	15.9	(27)	
		MCDB	16.8	18.3	23.9	26.1	29.3	30.8	32.1	32.0	30.7	27.8	21.2	17.8	(28)	
	2%	WB	10.9	12.3	16.5	18.5	22.7	25.3	26.9	26.2	24.5	21.5	15.8	12.6	(29)	
		MCDB	12.7	14.5	20.7	23.7	27.8	30.5	31.3	30.9	29.5	25.3	19.0	14.0	(30)	
	5%	WB	7.7	9.1	15.0	17.3	21.6	24.4	25.9	25.2	23.5	19.7	13.9	9.2	(31)	
		MCDB	9.8	12.5	18.5	22.5	26.2	29.3	30.6	30.0	28.0	22.4	17.1	10.9	(32)	
	10%	WB	4.7	6.0	12.6	15.7	20.5	23.5	25.1	24.4	22.4	17.7	11.9	6.6	(33)	
		MCDB	7.5	9.1	15.9	20.7	24.8	28.1	29.8	28.6	26.4	20.4	15.0	8.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.1	10.1	11.2	10.3	10.2	10.7	10.8	12.1	11.1	9.7	7.9	(35)	
		MCDBR	12.4	13.7	13.1	14.4	11.8	11.5	11.7	12.8	13.5	14.0	13.1	12.0	(36)	
	5% WB	MCWBR	9.7	9.8	7.6	7.2	5.5	5.3	5.1	5.1	5.4	7.0	8.5	9.4	(37)	
		MCWBR	11.4	13.1	11.8	11.9	10.4	10.5	10.4	10.4	11.3	10.9	11.7	11.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.296	0.315	0.338	0.389	0.432	0.431	0.457	0.447	0.400	0.351	0.304	0.290	(40)	
		taud	2.459	2.393	2.415	2.266	2.211	2.258	2.175	2.213	2.305	2.446	2.522	2.495	(41)	
	Ebn at Noon		877	904	919	885	851	850	823	821	839	852	864	859	(42)	
		Edh at Noon	79	97	106	131	142	135	146	137	117	92	74	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.95	2.77	3.73	4.87	5.54	6.31	6.28	5.51	4.76	3.29	2.22	1.66	(44)	
	RadStd		0.22	0.27	0.28	0.33	0.43	0.54	0.39	0.32	0.30	0.37	0.20	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	+78

Nomenclature: See separate page